# Rational nonbranch point: principal ideal and actual DVR

Formal proof packet

28 October 2026

This packet records a checked local geometry theorem. It contains no experiment, target result, or cost ratio. The Lean source, compiler log, editable SVG, and CSV gate table accompany this PDF.

## Checked boundary

Let $K$ be any field, $f(X) = X^5 + a_3X^3 + a_2X^2 + a_1X + a_0$, and $(a, y) \in K^2$ satisfy $y^2 = f(a)$. Assume $f$ is separable and $2y \neq 0$. The point evaluation ideal in $K[X, Y]/(Y^2 - f(X))$ is $(X - a, Y - y)$. In the localization at this ideal, $Y + y$ is a unit because its residue is $2y$. The curve relation gives

$$(Y - y)(Y + y) = (X - a)q_a(X).$$

Thus the local maximal ideal is generated by $X - a$. This generator is nonzero, and the separability bridge already proves that the affine coordinate ring and localization are domains. The checked DVR criterion gives a discrete valuation ring. The checked ring equivalence transports this structure to the *actual weighted*-Proj *point stalk*.

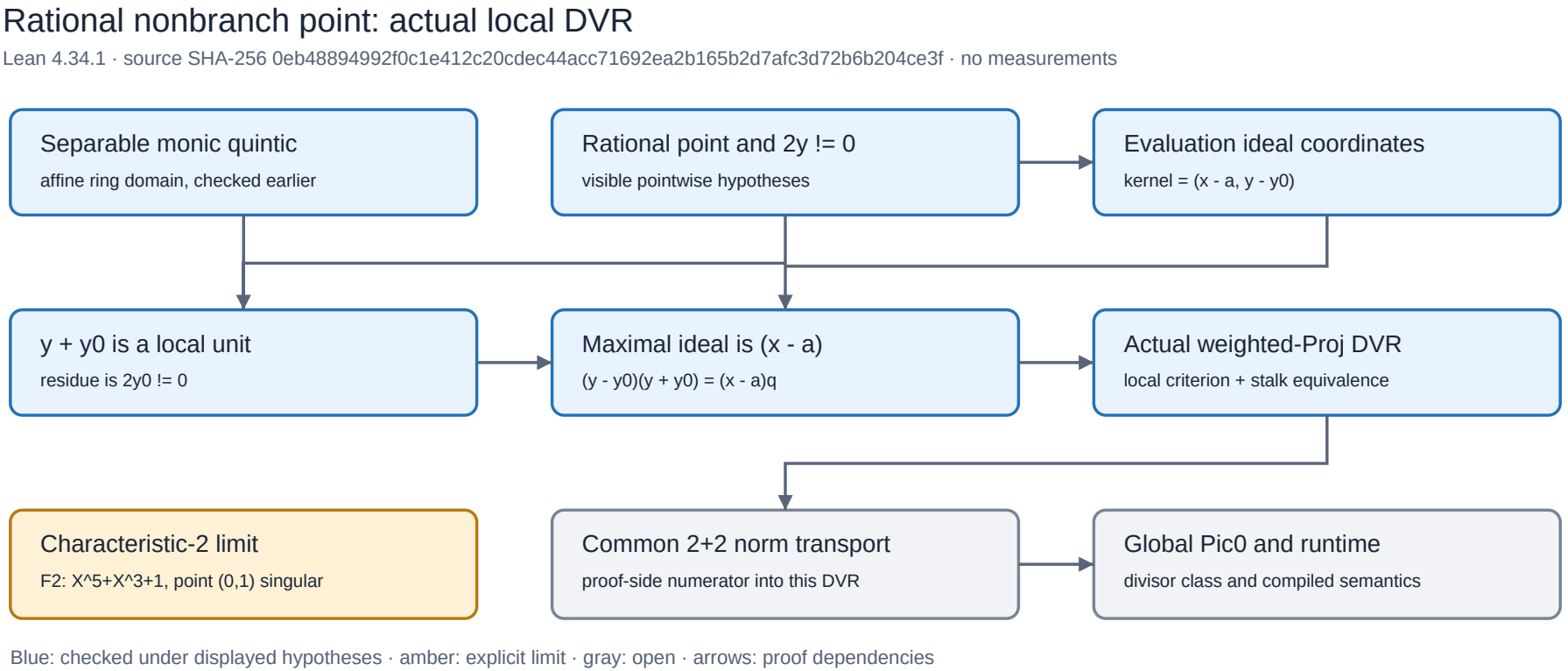
## Scope and verification

The condition $2y \neq 0$ is essential for this proof and cannot be replaced by $y \neq 0$ in characteristic two. Over $\mathbf{F}_2$, $f = X^5 + X^3 + 1$ is separable, yet $(0, 1)$ is on the curve and both partial derivatives vanish there. The alternate pointwise case $f'(a) \neq 0$ is not formalized here.

The direct pinned Lean 4.34.1 compile exited zero against Mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612`. Ten printed axiom audits list only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Lean source SHA-256: `0eb48894992f0c1e412c20cdec44acc71692ea2b165b2d7afc3d72b6b204ce3f`. The compiler log retains nonfatal style-linter suggestions. The local DVR criterion is the pinned Mathlib `IsDiscreteValuationRing.TFAE`, corresponding to Stacks Project Tag 00PD. Stalk transport uses the prior checked open-immersion equivalence, corresponding to Stacks Tag 01HD.

Formula-specific norm transport into this stalk remains open, as do global principal-divisor/$\mathrm{Pic}^0$ transport and compiled Rust semantics. The CSV gives the exact status of each theorem and gate.

## Proof dependencies and open gates




Blue nodes are checked under the displayed hypotheses. Amber records the characteristic-two limit; gray nodes are open. Arrows show proof dependencies, not measured performance.